

ZIN MIN KHANT (Alfred)

Senior Software Engineer · Backend Architecture · Identity & Security · AI-Enabled Systems

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SUMMARY

Senior software engineer with 7+ years across backend architecture, authentication and identity platforms, cloud infrastructure, blockchain, and AI integration. I led a re-build of a labor-rights platform's backend from Django to Node/TypeScript while it served 500,000+ workers, built a centralized OIDC single-sign-on gateway used by three products, and designed and shipped a blockchain-and-AI compliance SaaS on my own. I work comfortably across the stack, from Python, TypeScript, Node, and C#/.NET on the backend to React, Next.js, and Vue on the frontend, with a consistent focus on security and scalability.

TECHNICAL SKILLS

Languages: TypeScript, Python, JavaScript, C#/.NET, SQL (T-SQL), Solidity, PHP

Backend: Node.js/Express, FastAPI, Django/DRF, .NET/Entity Framework, Laravel, Celery, REST & GraphQL APIs

Frontend: React (TypeScript), Next.js, Vue.js, Vite, TanStack, Redux, Tailwind/SCSS, offline-first (Dexie/IndexedDB)

Data: PostgreSQL, MySQL, Microsoft SQL Server, Oracle, Redis, MongoDB, SQLAlchemy, Sequelize

Identity & Security: OAuth2/OIDC, WorkOS, RS256 JWT/JWKS, SAML SSO, AES-256-GCM, HMAC, multi-tenant RBAC

Cloud & DevOps: AWS (ECS Fargate, Lambda, S3, CDK, Secrets Manager), GCP (App Engine, Compute), Docker, Kubernetes, GitHub Actions CI/CD, Cloudflare

AI: Multi-provider LLM integration (Anthropic, OpenAI), embeddings and entity resolution, PII redaction before model calls

Blockchain: Solidity smart contracts, web3.py / Web3.js, Ethereum and Polygon, on-chain anchoring, wallet integration (MetaMask)

Architecture & Practice: Distributed systems, legacy modernization (strangler-fig), multi-database persistence, observability (OpenTelemetry/Prometheus), Agile/Scrum, technical leadership and mentoring

EXPERIENCE

Full Stack Developer (Tech & Innovation), [Issara Institute](#)

2025 – Present

Bangkok, Thailand · Architecture, Identity & Platform Engineering

I lead backend, identity, and security work on the Tech and Innovation team, building and modernizing a multi-tenant labor-rights platform for worker protection and ethical supply chains. The platform covers **10,000+ suppliers and 500,000+ workers** across Southeast Asia.

- **Led the re-platforming** of the analytics product's backend from Python/Django to **Node.js and TypeScript** (Express, Sequelize, Zod). I used a **strangler-fig** approach so we avoided a high-risk big-bang rewrite and shipped without downtime.
- Designed a **polyglot, fault-tolerant persistence layer**: a primary application database alongside read-only connections to two legacy MySQL systems. The legacy connections degrade gracefully, so new features keep working even when a legacy source is unavailable.
- Built a **centralized OIDC single-sign-on and identity-federation gateway** (Flask and WorkOS, deployed as a stateless AWS Lambda with CDK) that gives three products one login. It correlates identities across products through WorkOS metadata rather than a central user database, and supports centralized session revocation.
- Owned the **authentication and security** layer: RS256/JWKS verification with key rotation and issuer/expiry validation, **AES-256-GCM response encryption**, refresh-token rotation, and privilege-escalation hardening. I also ran a **zero-downtime migration** from legacy JWT-2FA to WorkOS SSO using a dual-provider backend.
- Built the analytics SPA in **React 19, Vite, and TypeScript** with a TanStack data layer, **offline-first sync (Dexie/IndexedDB)**, role-based multi-tenant access, feature flags, i18n, and geospatial dashboards, covered by Vitest and Playwright tests.
- Added **AI to the data layer** with an embeddings-based pipeline that reconciles supplier and recruiter identities across the legacy and new systems.
- Set technical direction on a **three-person engineering team**: I run **design reviews, mentor and review** the other developers' work, and translate architecture decisions for non-engineering stakeholders across the products that depend on the platform.

Tech: TypeScript, Node.js, Express, Django/DRF, Python, React, Vite, MySQL, WorkOS/OIDC, Flask, AWS Lambda/CDK, Docker, GitHub Actions.

Technical Lead (Freelance), **Booppa Smart Care LLC**

Dec 2025 – Jun 2026

Singapore (Remote)

As the sole technical lead, I architected and built a production multi-tenant **compliance-evidence SaaS** spanning **58 FastAPI backend services**, a **139-route Next.js 14 App Router frontend**, a Polygon smart contract, AI classification, Stripe payments, and AWS cloud infrastructure.

- Designed a **tamper-evidence system** that combines a deterministic SHA-256 content hash, a cryptographically **hash-chained audit log**, and **on-chain anchoring** through a custom EvidenceAnchorV3 Polygon smart contract (idempotent single and batch anchoring). A buyer can verify the evidence independently, without trusting Booppa.
- Built a **provider-agnostic AI classification layer** (Anthropic, OpenAI, DeepSeek, Ollama) for PII/NRIC and privacy-policy analysis. Sensitive data is **redacted before any external model call**, and a heuristic fallback runs with no external calls at all.
- Engineered a **race-resistant Stripe checkout-to-fulfilment pipeline** across one-time and subscription products and several buyer and vendor lanes, running on **asynchronous Celery/Redis** workers for scanning, PDF generation, S3 upload, and anchoring.
- Delivered a hardened **Next.js 14** frontend with three middleware-enforced auth zones, **HMAC-signed paywall cookies**, JWT-refresh rotation, and real-time notifications. It runs on **AWS ECS Fargate behind a Cloudflare Tunnel**, with SAML SSO and OpenTelemetry/Prometheus observability.

Tech: FastAPI, PostgreSQL, Celery/Redis, web3.py/Solidity (Polygon), Next.js 14, TypeScript, Stripe, AWS ECS/Cloudflare.

Blockchain / Full-Stack Developer (Freelance)

2024 – 2025

Remote (Germany-based company, Thai client)

- Designed and deployed **Ethereum smart contracts in Solidity** for token transactions and on-chain data validation, with a **Vue.js** front end and **Node.js** services connected through **Web3.js**. Added **MetaMask** wallet integration for authentication and transaction signing.

Tech: Solidity, Ethereum, Vue.js, Node.js, Web3.js, MetaMask.

Senior Full Stack Developer, **Wunderfauks**

Dec 2023 – Dec 2024

Singapore

Led a **team of three junior developers** delivering client web products for the Singapore agency.

- Built "**RWC Game**," a **gamified rewards platform** for Real World Cruise, where passengers play games to earn coins and redeem rewards. It is a stateful product (player accounts, a coin economy, redemptions) on a WordPress backend, with **two custom PHP plugins and a JWT-secured REST API** for the gameplay client. I also built a feedback and review platform for **Singapore Post (Singpost)** on a custom WordPress theme.
- Built a **decoupled headless-CMS platform** (M-Vista) that runs WordPress as a structured content API behind a separate **Next.js** front end, plus an e-commerce site with secure payment gateways, multilingual sites, and custom theme and plugin work for corporate clients (Nextevo x2, Orange Clove).

Tech: WordPress/PHP (custom plugins), JWT, Next.js, React, headless CMS, multilingual (i18n), payment gateways.

Senior Full Stack Developer, **SCW Myanmar**

2020 – 2023

Myanmar

- Built customer-facing web applications from design mock-ups and specifications, including a launched **e-commerce platform**, a **travel and tour site**, and a **hotel website with an integrated booking system**.
- Named **Employee of the Year in 2021 and 2022** for delivery and impact.

Web Developer (Freelance)

2018 – 2020

Remote

- Built client web applications including a **property-listing platform with an admin panel and multi-role access control**; gathered requirements and managed content for internal and external sites.

EDUCATION

B.Sc. (Hons) Information Technology for Business, Oxford Brookes University (UK), *Upper Second Class Honours* · **Diploma in Business IT**, NCC Education (Strategy First University) · **CCNA**, Cisco · **IGCSE O Levels**, ILBC

AWARDS

Employee of the Year (2021 & 2022), SCW Myanmar · **Runner-Up**, Deloitte Risk Intelligence Challenge (2019) · **Honorable Mention**, Myanmar Youth Technopreneurship (2019)

LANGUAGES

English (Full Professional Proficiency) · Burmese (Native) · Thai (Basic)